

Finite Element Analysis of an Aircraft Fuselage Section

CAE Advanced Applications Assignment



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1 Introduction

This report presents the finite element analysis (FEA) of a representative aircraft fuselage section, carried out as part of the CAE Advanced Applications module at Cranfield University [1, 2, 3]. The objective is to determine the stress distribution and maximum deflection within the structural components of the fuselage under representative operational loading conditions, and to demonstrate competence in the practical application of a commercial finite element code (Abaqus/CAE).

The fuselage is a semi-monocoque structure comprising a thin outer skin, longitudinal stringers, circumferential ring frames, a passenger floor, and floor stiffeners. Each component plays a distinct structural role:

- The **skin** carries shear loads and resists cabin pressurisation;
- The **stringers** carry longitudinal bending loads and stabilise the skin against buckling;
- The **ring frames** maintain the cross-sectional shape under pressure loads and transfer loads between components;
- The **floor panel and stiffeners** support passenger and cargo loads and distribute them to the frames.

The report is structured as follows. Section 2 provides the theoretical background of the finite element formulations employed, namely shell and beam elements, discussing their degrees of freedom, stress output locations, and accuracy considerations. Section 3 details the model generation process, including geometry creation, partitioning, material assignment, meshing strategy, boundary conditions, and applied loads. Section 4 presents the simulation results for the baseline model, including deformed shapes, stress distributions, and a

mesh sensitivity study. Section 5 explores two alternative configurations by varying the floor loading and cabin pressure independently. Section 6 discusses engineering design improvements to reduce floor deflection. Finally, Section 7 summarises the findings and provides recommendations.

Statement on the Use of Generative AI

During the preparation of this work, the author used ChatGPT (OpenAI) and GitHub Copilot to assist with:

- $\text{\LaTeX}$ formatting and report structure organisation
- Technical writing refinement and clarity improvement

After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the final report's content, analysis, and results.

2 Element Formulations and Theoretical Background

This section describes the finite element formulations selected for the fuselage model and provides a theoretical justification for the modelling choices. The fuselage structure is idealised using two principal element types: **shell elements** for the skin and floor panels, and **beam elements** for the stringers, ring frames, and floor stiffeners.

2.1 Shell Elements (S4R)

2.1.1 Element Definition and Key Features

Shell elements are used to represent thin-walled structural components where one dimension (the thickness) is significantly smaller than the other two. In this model, the **S4R** element (4-node, reduced-integration, general-purpose shell) from the Abaqus element library is employed for the fuselage skin and floor panels [4].

Key characteristics of the S4R element:

- **Nodes:** 4 corner nodes, each with 6 degrees of freedom (3 translational: u_x, u_y, u_z ; and 3 rotational: $\theta_x, \theta_y, \theta_z$);
- **Integration:** Reduced integration with a single integration point in the element plane (hourglass control is applied to prevent spurious zero-energy modes);
- **Thickness:** Defined through section properties (not modelled explicitly in the geometry), allowing the element to account for bending stiffness via through-thickness integration points (by default 5 Simpson integration points through the thickness);
- **Surface orientation:** The element normal direction determines the top (SPOS) and bottom (SNEG) surfaces. This is critical for applying pressure loads and interpreting

stress output.

2.1.2 Stress Output for Shell Elements

Shell elements provide stress results at multiple locations:

- **Section points:** Stresses are computed at each integration point through the thickness. The outermost section points (top and bottom surfaces) provide the maximum bending stresses;
- **In-plane stresses:** σ_{11} , σ_{22} , and τ_{12} in the local element coordinate system, where direction 1 and 2 lie in the shell surface;
- **Mises stress:** The von Mises equivalent stress is computed from the full stress tensor and is used to assess yielding under multi-axial loading;
- **Output locations:** Stresses can be requested at integration points, element centroids, or extrapolated to nodes. The most accurate stresses are at the **integration points**; nodal values are obtained by extrapolation and averaging, which can introduce smoothing errors [5].

For the purposes of this analysis, the von Mises stress at the outermost section points is the primary metric for evaluating structural adequacy.

2.1.3 Stress Calculation in Shells

Stresses in shell elements arise from the combination of membrane (in-plane) and bending (out-of-plane) actions. For a thin shell under combined loading:

$$\sigma_{total} = \sigma_{membrane} \pm \sigma_{bending} \tag{1}$$

where the membrane stress is uniform through the thickness and the bending stress varies linearly, reaching its maximum at the outer fibres. The most accurate stresses in Abaqus shell elements are obtained at the reduced integration point (element centroid in the plane), where they are computed directly from the constitutive equations without extrapolation [4].

2.2 Beam Elements (B31)

2.2.1 Element Definition and Key Features

Beam elements are used to represent slender structural members where the cross-sectional dimensions are small compared to the length. In this model, the **B31** element (2-node linear Timoshenko beam) is used for the stringers, ring frames, and floor stiffeners [4].

Key characteristics of the B31 element:

- **Nodes:** 2 end nodes, each with 6 degrees of freedom (3 translational: u_x, u_y, u_z ; and 3 rotational: $\theta_x, \theta_y, \theta_z$);
- **Formulation:** Timoshenko beam theory, which accounts for both flexural deformation and transverse shear deformation. This is appropriate for beams with moderate length-to-depth ratios;
- **Cross-section profile:** Defined as a separate beam section property (e.g., I-section, box, or general profile). The profile dimensions define the area, moments of inertia, and shear area;
- **Orientation:** A beam section orientation vector ($\mathbf{n}_1$) must be specified to define the local axes of the cross-section. This is critical for correctly orienting asymmetric profiles (such as Z or L sections) and for interpreting bending stress results about the correct axes.

2.2.2 Stress Output for Beam Elements

Beam elements provide stress results at specific section points across the profile:

- **Section points:** Stresses are output at predefined locations across the cross-section (typically at the extremities of the flanges and web). The number and location of section points depend on the profile type;
- **Axial stress (σ_{11}):** The dominant stress component, arising from a combination of axial force and bending moments about both principal axes:

$$\sigma_{11} = \frac{N}{A} + \frac{M_y \cdot z}{I_y} + \frac{M_z \cdot y}{I_z} \quad (2)$$

where N is the axial force, A the cross-sectional area, M_y and M_z are bending moments, and I_y , I_z are the second moments of area;

- **Shear stresses:** Transverse shear stresses (τ_{12} , τ_{13}) are also available for Timoshenko beams;
- **Output locations:** Stresses are evaluated at integration stations along the beam length and at section points across the profile.

2.2.3 Stress Calculation in Beams

The axial stress in a beam element is the superposition of direct stress (due to axial load) and bending stress (due to bending moments). The maximum stress occurs at the section point furthest from the neutral axis. In Abaqus, the beam section force output (SF) provides the internal forces and moments, which can be used to verify the computed stresses.

2.3 Comparison: Shell vs Beam Elements

Feature	Shell (S4R)	Beam (B31)
Nodes per element	4	2
DoF per node	6 (3 trans. + 3 rot.)	6 (3 trans. + 3 rot.)
Geometry representation	Mid-surface	Centreline (1D)
Cross-section	Thickness (through section property)	Profile (I, box, general, etc.)
In-plane stresses	$\sigma_{11}, \sigma_{22}, \tau_{12}$	σ_{11} (axial only)
Shear deformation	Through-thickness (Mindlin theory)	Transverse (Timoshenko theory)
Bending stress	Linear through thickness	Linear across section height
Ideal for	Skin, floor panels (thin plates)	Stringers, frames, stiffeners (slender members)

Table 1: Comparison of shell and beam element formulations

2.4 Justification of Modelling Choices

The mixed shell–beam modelling approach is the standard idealisation for aircraft fuselage analysis [6]. Using shell elements for the skin and floor captures the bi-axial membrane stresses due to pressurisation and the bending response under floor loading. Using beam elements for the stringers and frames is computationally efficient and appropriate given their slender geometry.

An important consideration in this mixed-element approach is the **offset** of beam elements. The stringers and frames are physically attached to the skin surface, not at the skin mid-plane. In Abaqus, beam offsets can be used to position the beam reference line at the correct distance from the shell mid-surface. This ensures that:

1. The eccentricity between the beam neutral axis and the shell mid-surface is correctly captured;
2. The bending stiffness coupling between the skin and stiffeners is accurately represented;
3. The overall structural response (deflections and stress distribution) is realistic.

The unit system adopted throughout this analysis is the SI system: lengths in metres (m), forces in Newtons (N), stresses in Pascals (Pa), and mass density in kg/m^3 .

3 Model Generation

This section describes the step-by-step process used to create the finite element model of the fuselage section in Abaqus/CAE [7, 8, 9, 10]. All dimensions, material properties, and loading data are taken from the assignment specification.

3.1 Geometry Creation

3.1.1 Part Definition – Sketch and Base Extrusion

The fuselage cross-section was created as a **3D deformable shell** part. The initial sketch defines the circular cross-section of the fuselage skin, including the flat floor panel at the appropriate vertical position. The sketch is shown in Figure 1.

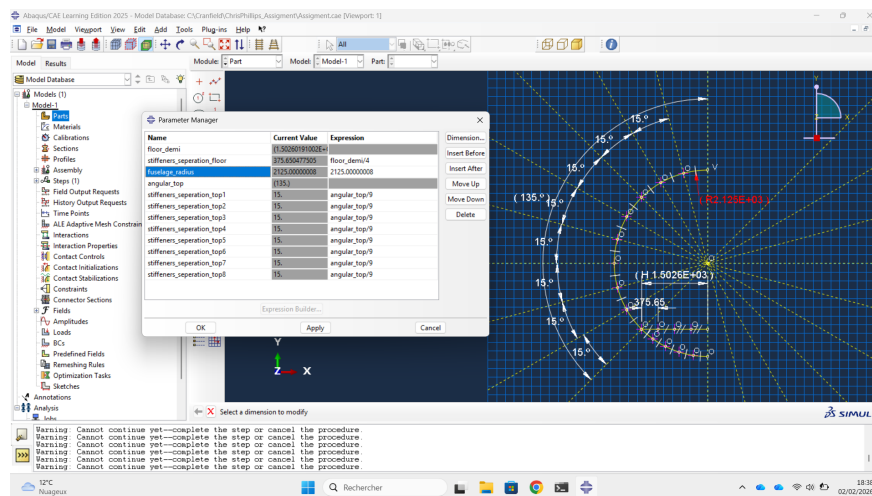


Figure 1: Part sketch defining the fuselage cross-section

The sketch was then extruded along the longitudinal axis to create the full-length fuselage barrel section. The base extrusion is shown in Figure 2.

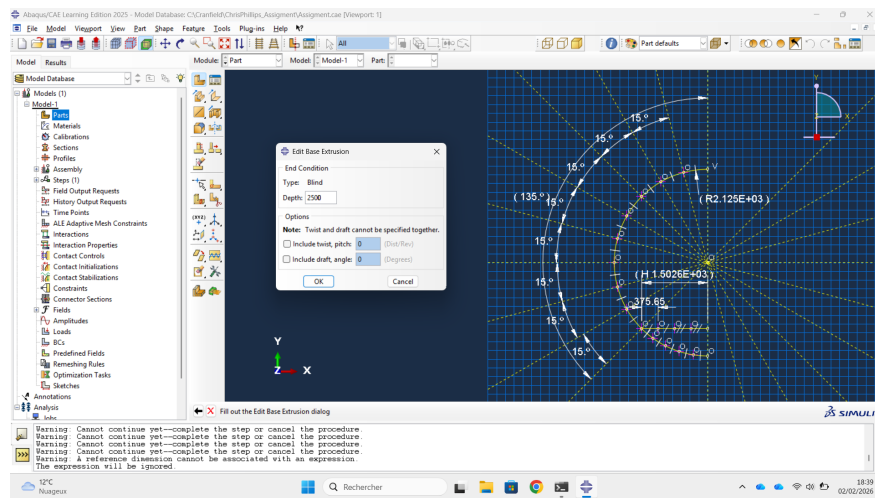


Figure 2: Base extrusion of the fuselage section

3.1.2 Partitioning for Ring Frames and Stringers

To position the ring frames and stringers at the correct locations, datum planes were created along the length and around the circumference of the fuselage. The datum planes for ring frame locations are shown in Figure 3.

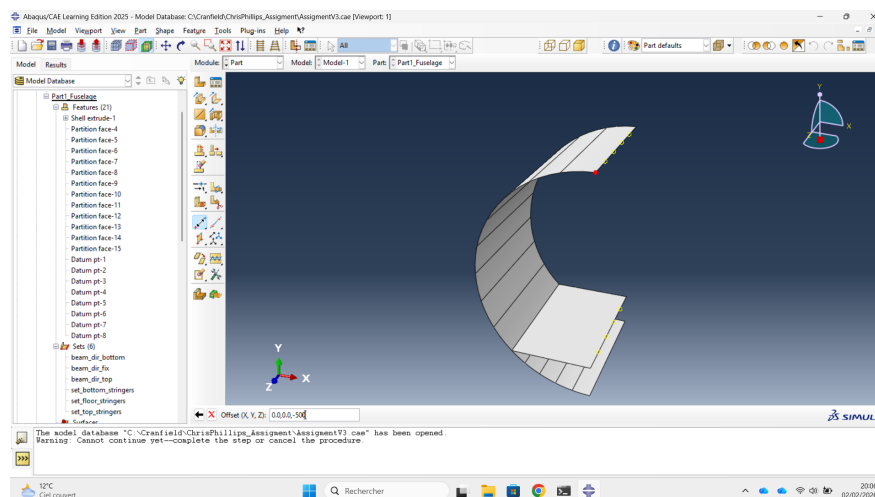


Figure 3: Datum planes for ring frame positioning

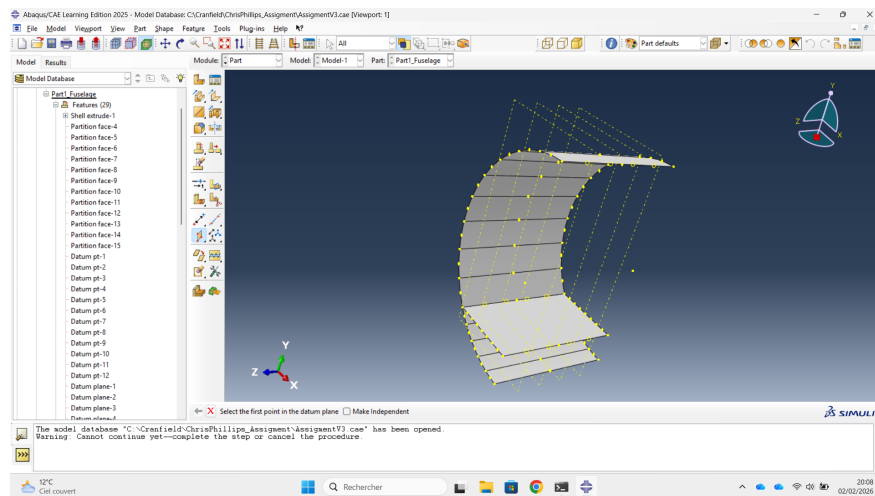


Figure 4: Creation of additional reference planes

The part was then partitioned at each datum plane location. Partitioning serves two essential purposes: (1) it creates edges to which beam elements (stringers, frames, stiffeners) can be assigned, and (2) it provides mesh control by creating structured regions for a high-quality mesh. The partitioned geometry is shown in Figure 5.

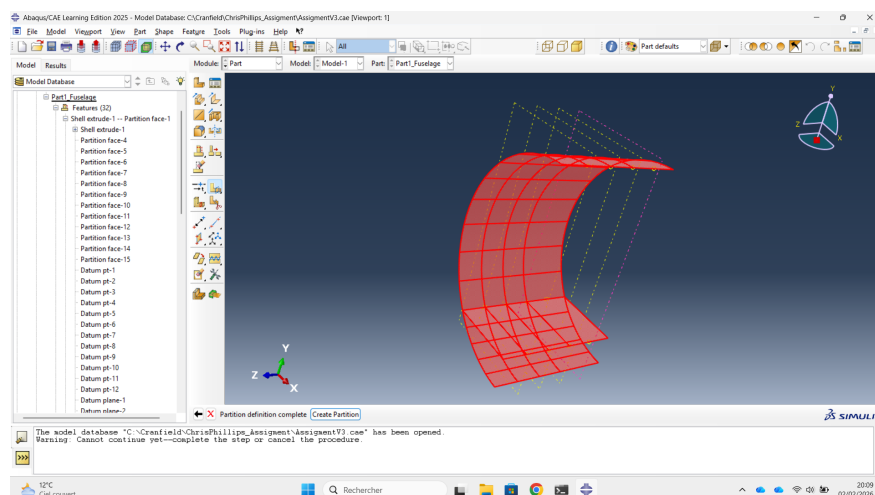


Figure 5: Partitions created at each datum plane location

3.1.3 Sets and Regions

After partitioning, geometry sets were created for each structural sub-component: **Stringers**, **Ring Frames**, and **Floor Stiffeners**. Organising the geometry into sets is critical for:

- Assigning different section properties to different structural components;
- Applying specific element types to each region;
- Extracting results for individual components during post-processing.

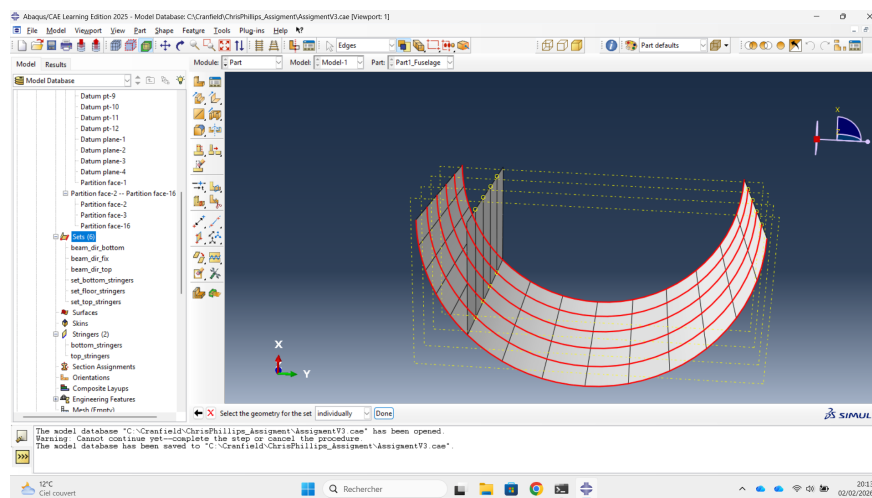


Figure 6: Geometry sets defined for each structural sub-component

The overall appearance of the partitioned part is shown in Figure 7.

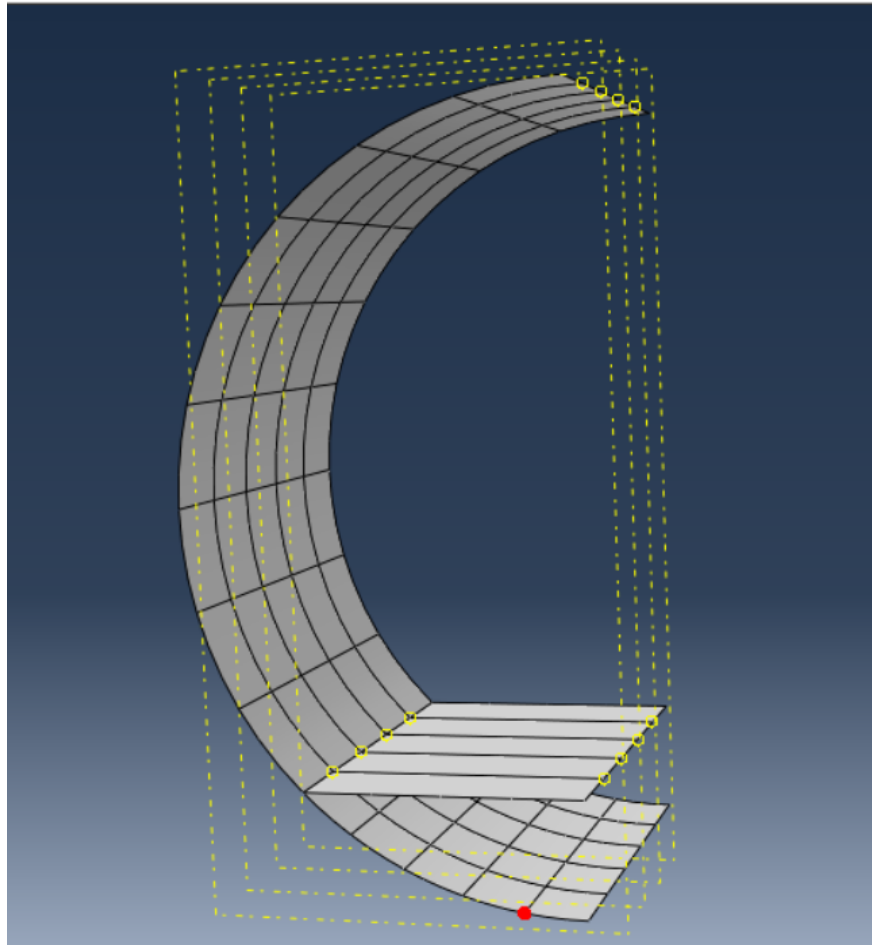


Figure 7: Overview of the partitioned fuselage part

3.2 Material Properties

Three distinct materials were defined to represent the different structural components of the fuselage. All material properties were entered as specified in the assignment data sheet. The material property verification is shown in Figure 8.

Component	Material	Density [kg/mm ³]	Young's Modulus [GPa]	Poisson's ratio [-]	Yield Stress [Mpa]
Skin	AA 2090-T83	2.64×10^{-6}	79GPa	0.33	455
Skin Frame	AA 2199	2.64×10^{-6}	77.2GPa	0.33	450
Stringer	AA 7075-T6	2.58×10^{-6}	79GPa	0.30	503
Floor	AA 7076-T3	2.70×10^{-6}	80GPa	0.33	450
Floor Stiffeners	AA 2090-T83	2.64×10^{-6}	79GPa	0.33	455

Figure 8: Verification of material definitions in Abaqus

3.2.1 Material 1: Skin and Floor Panel

This material is applied to the fuselage skin and floor panel shell regions.

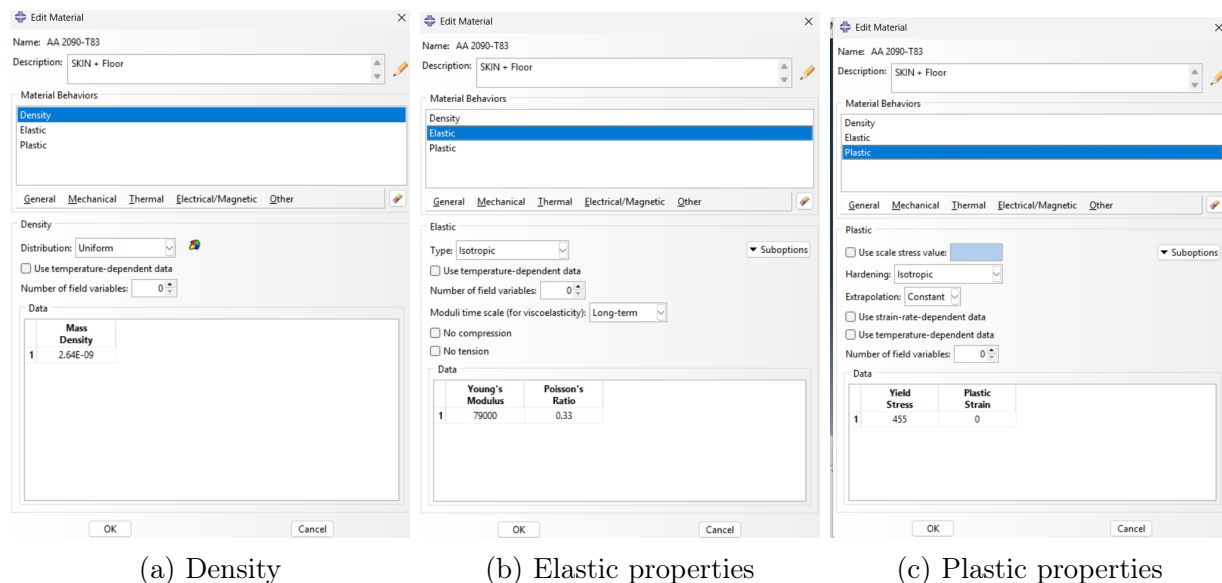


Figure 9: Material properties for Skin and Floor panel

3.2.2 Material 2: Skin and Ring Frame

This material is applied to the ring frame regions (and potentially the skin in frame-adjacent zones, depending on design variations).

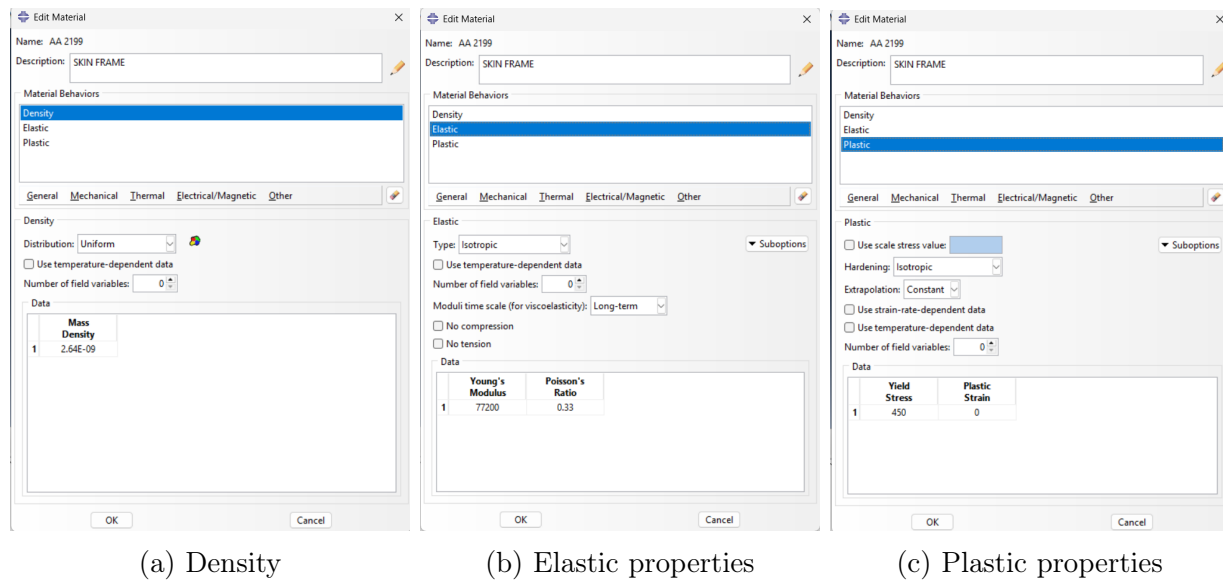


Figure 10: Material properties for Skin/Ring Frame

3.2.3 Material 3: Stringers

This material is applied to the longitudinal stringers.

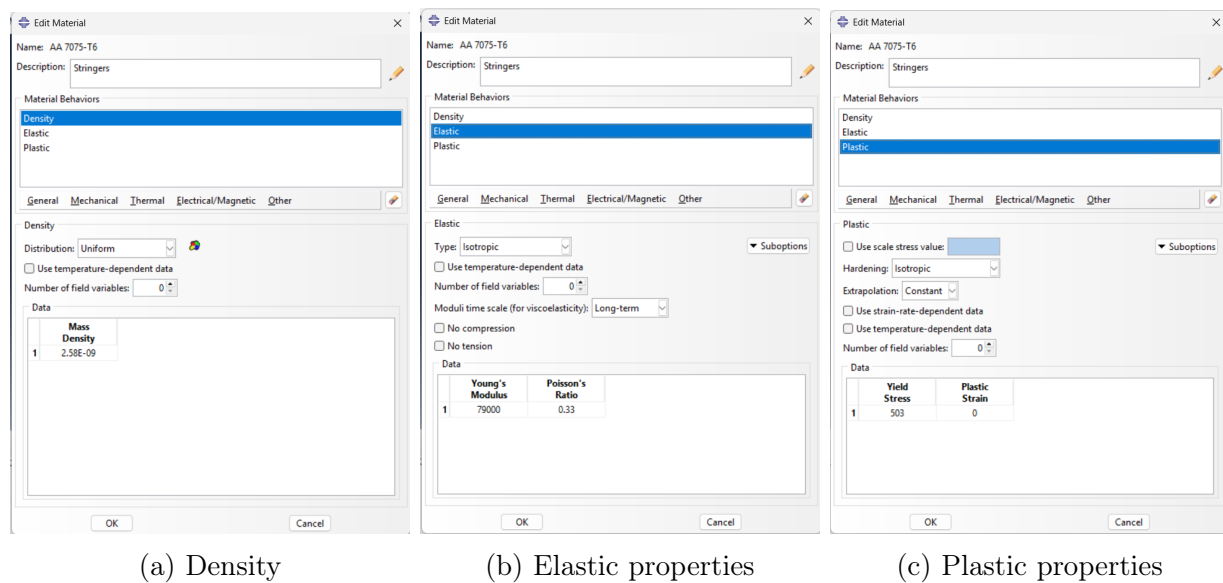
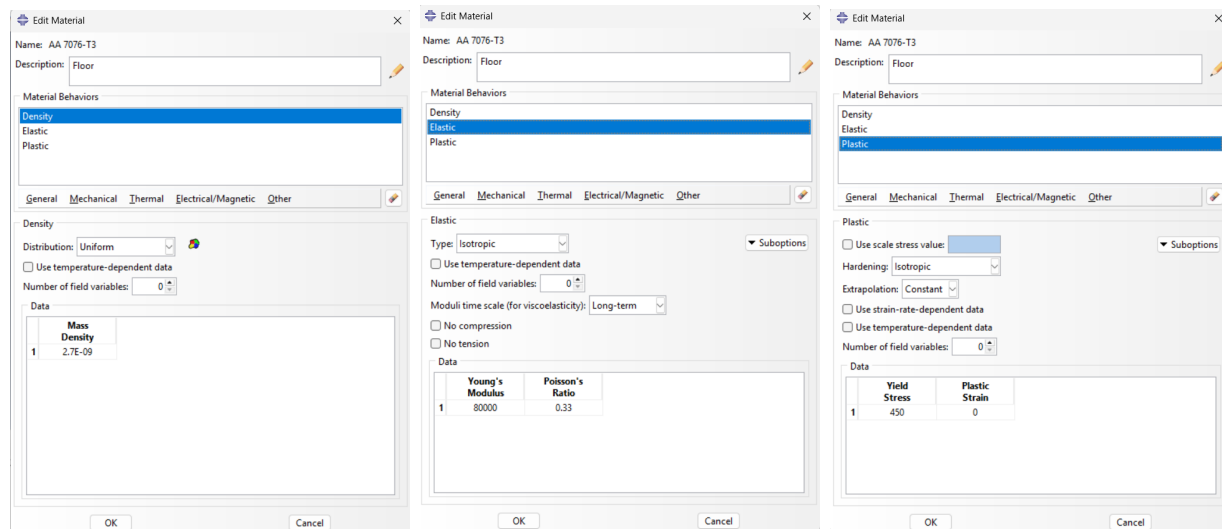


Figure 11: Material properties for Stringers

Additional material property screenshots from the Abaqus property module are shown below for completeness:



(a) Material summary 1

(b) Material summary 2

(c) Material summary 3

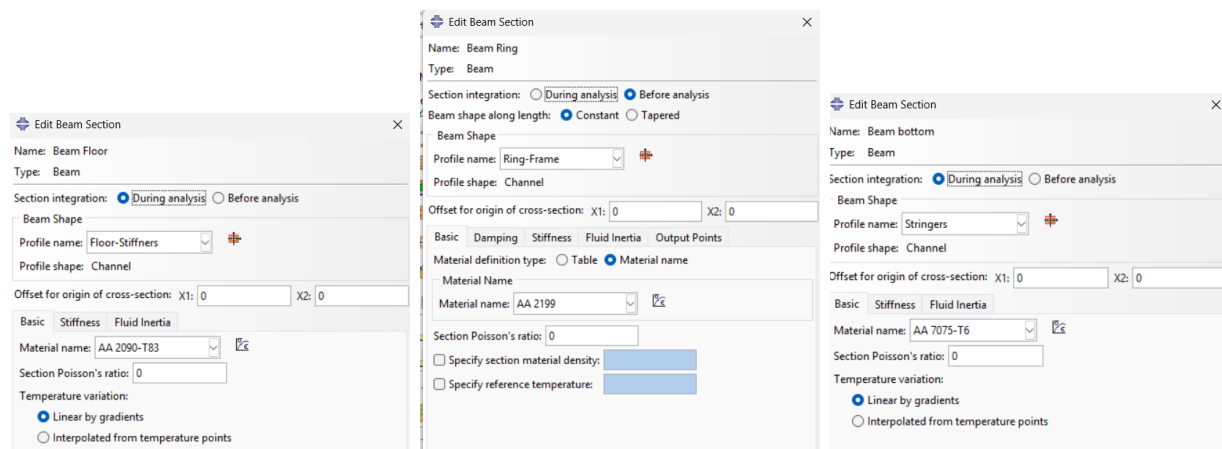
Figure 12: Additional material property verification

3.3 Section Properties

Section properties were assigned to each structural component to define the element cross-sectional behaviour.

3.3.1 Beam Sections

Beam section profiles were defined for the floor stiffeners, ring frames, and stringers. The cross-sectional profile shapes (I-section, Z-section, etc.) and their dimensions were taken from the assignment specification.

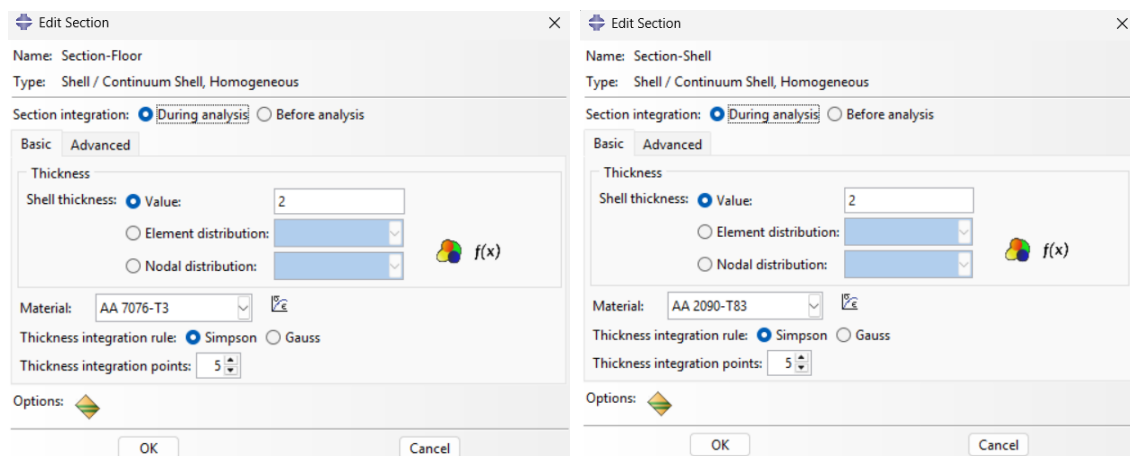


(a) Floor stiffener beam section (b) Ring frame beam section (c) Stringer beam section

Figure 13: Beam section definitions for structural sub-components

3.3.2 Shell Sections (Thickness Assignment)

The shell thickness was assigned to the skin and floor panel regions through shell section definitions.



(a) Floor panel thickness

(b) Skin thickness

Figure 14: Shell section thickness assignments

3.3.3 Beam Profile Definitions

The cross-sectional profiles used for each beam element component are presented below.

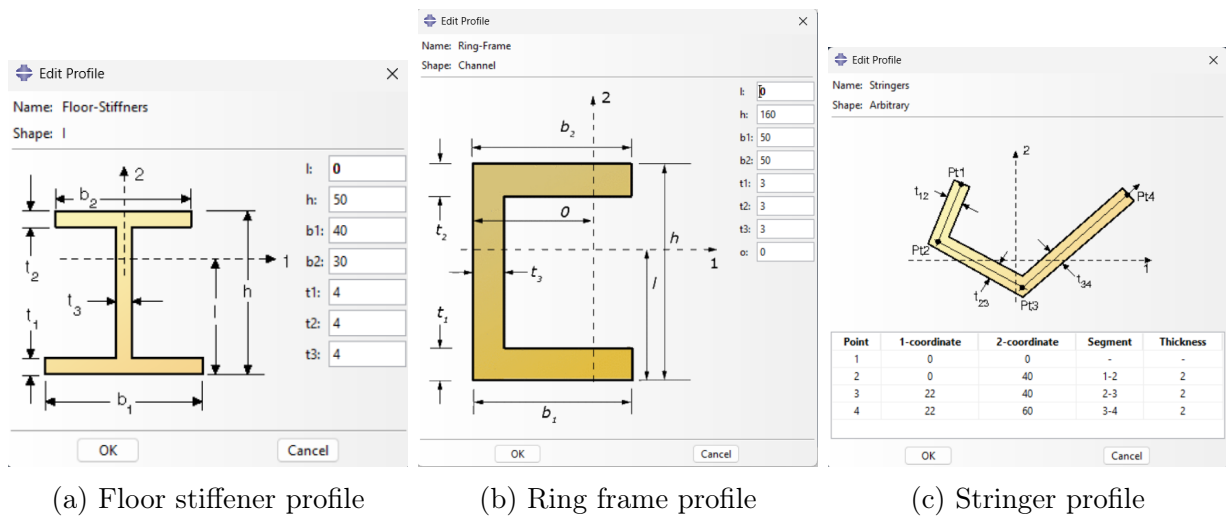


Figure 15: Beam cross-sectional profiles

The verification of section assignments in the model is shown in Figure 16.

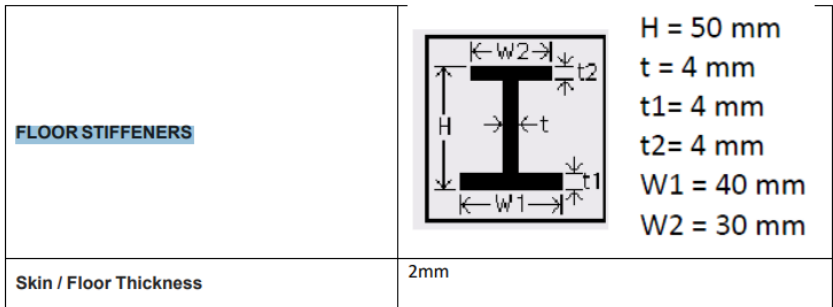


Figure 16: Verification of section assignments

A rendered view of the model showing the assigned sections (with beam profiles displayed) is presented in Figure 17.

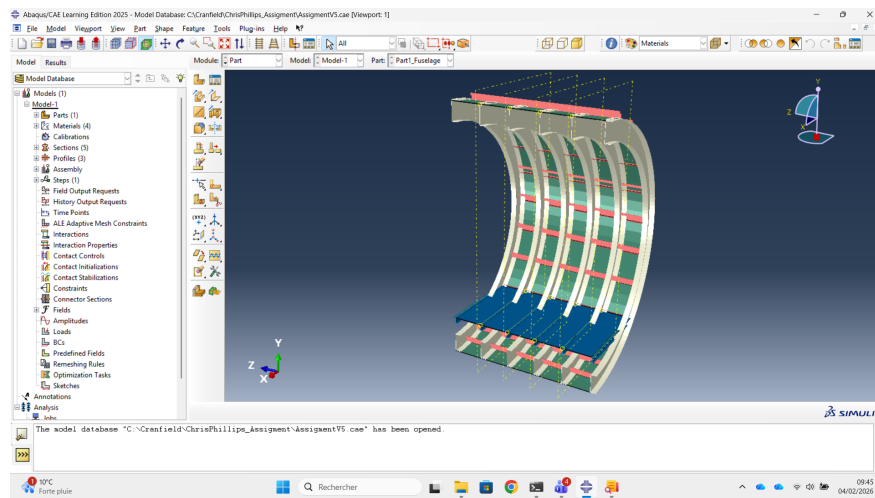


Figure 17: Rendered view of the model with section properties displayed

3.4 Element Type Assignment

The element types were carefully selected and assigned to each region of the model:

- **Skin and floor panels:** S4R (4-node reduced integration shell elements), as shown in Figure 18;
- **Stringers, ring frames, and floor stiffeners:** B31 (2-node linear Timoshenko beam elements), as shown in Figure 19.

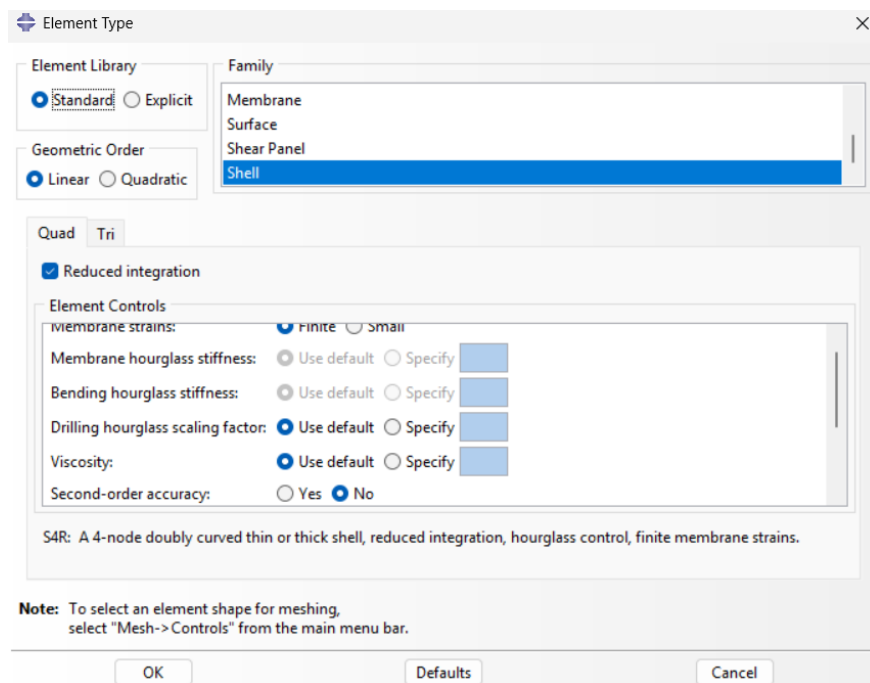


Figure 18: Shell element type assignment (S4R)

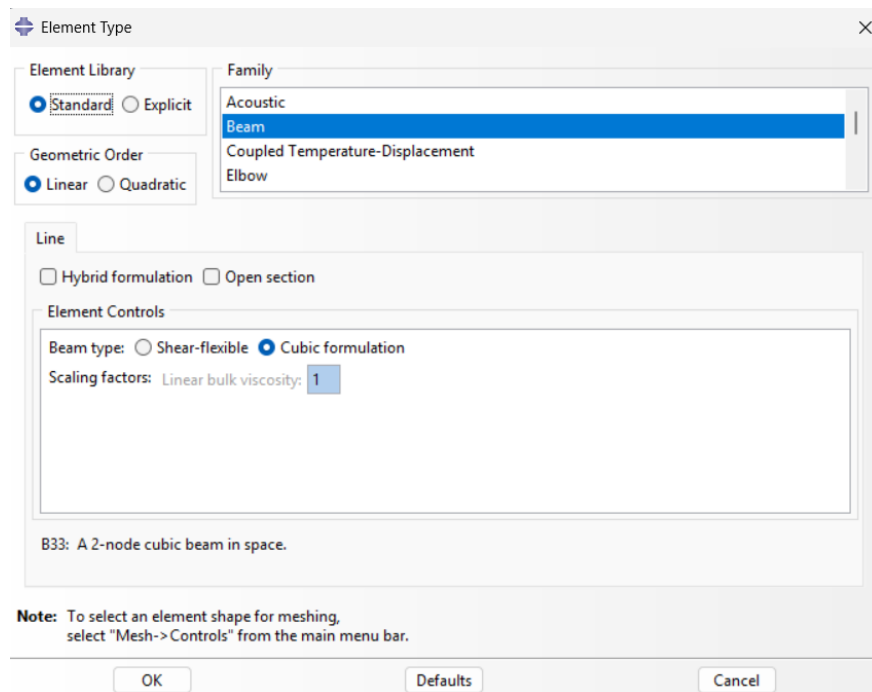


Figure 19: Beam element type assignment (B31) for stringers

3.5 Assembly and Step Definition

The part was instanced into the assembly module. Since the fuselage cross-section possesses a plane of symmetry along the vertical centreline, only **half of the fuselage** was modelled, with appropriate symmetry boundary conditions applied (see Section 3.6). This approach halves the computational cost whilst yielding identical results.

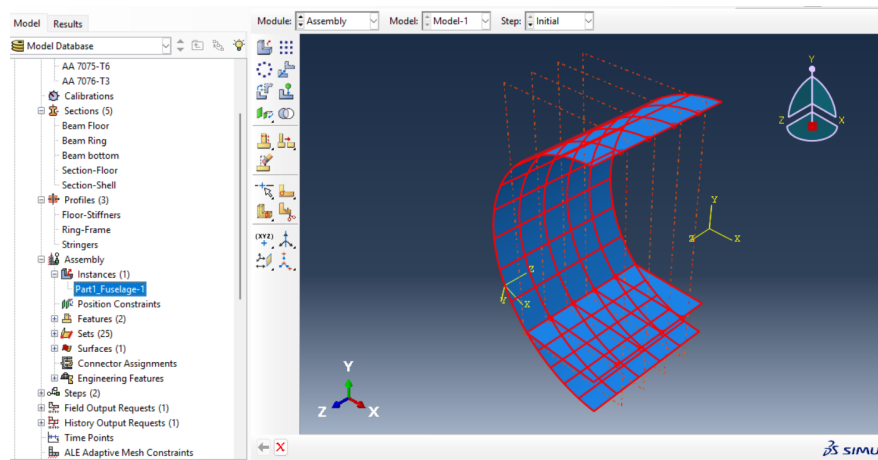


Figure 20: Assembly instance creation

A single **Static, General** analysis step was created to solve the linear static equilibrium under the applied loads. The step configuration is shown in Figure 21.

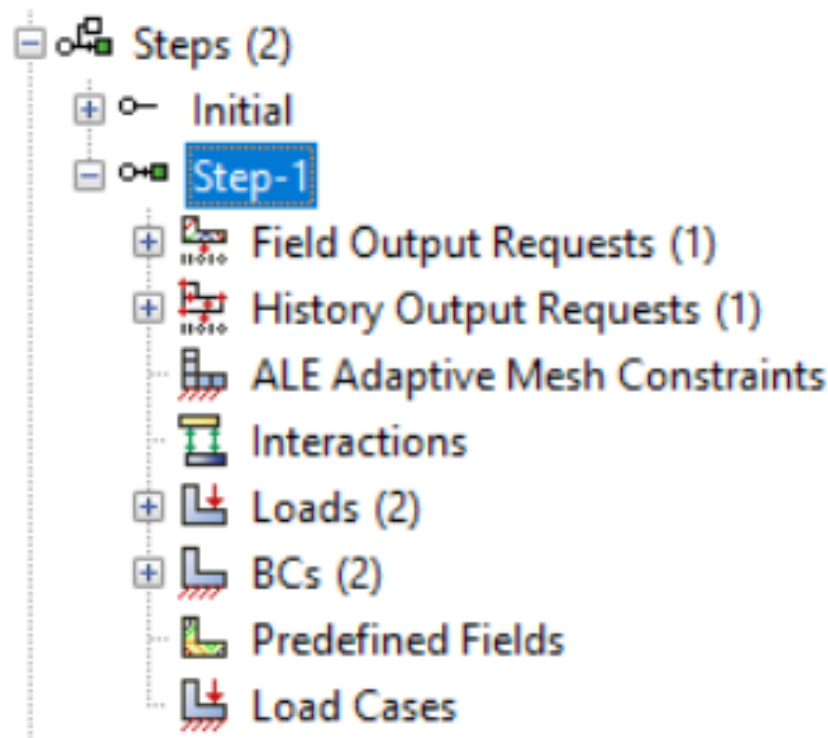


Figure 21: Step definition (Static, General)

3.6 Boundary Conditions

Two sets of boundary conditions were applied to the model:

3.6.1 Symmetry Boundary Condition

A **symmetry boundary condition** (ZSYMM or XSYMM, depending on the symmetry plane orientation) was applied along the vertical plane of symmetry. This constrains the out-of-plane translation and the two in-plane rotations at the symmetry face, as shown in Figure 22.

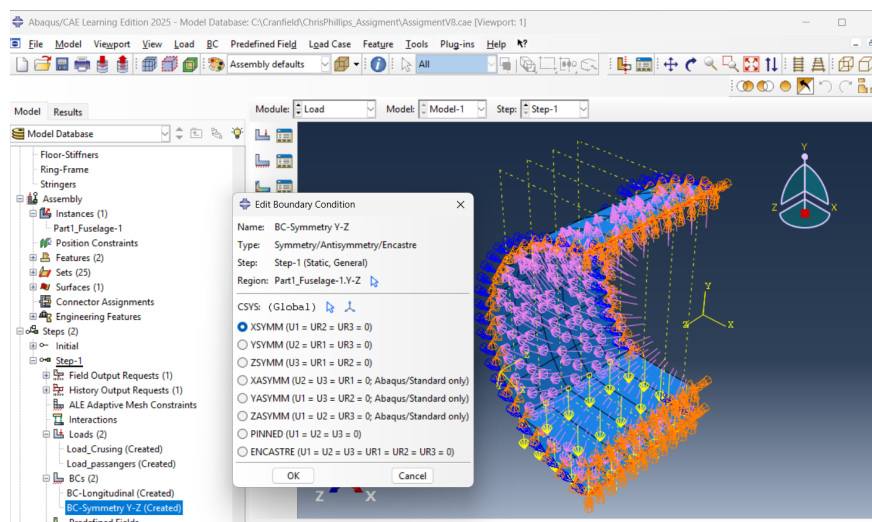


Figure 22: Symmetry boundary condition applied at the plane of symmetry

3.6.2 Longitudinal Restraint

At the far end of the fuselage barrel (opposite to the free end), a **longitudinal restraint** was applied to prevent rigid body motion along the fuselage axis whilst still allowing the section to expand radially under pressurisation. This simulates the support provided by the adjacent fuselage structure. The longitudinal boundary condition is shown in Figure 23.

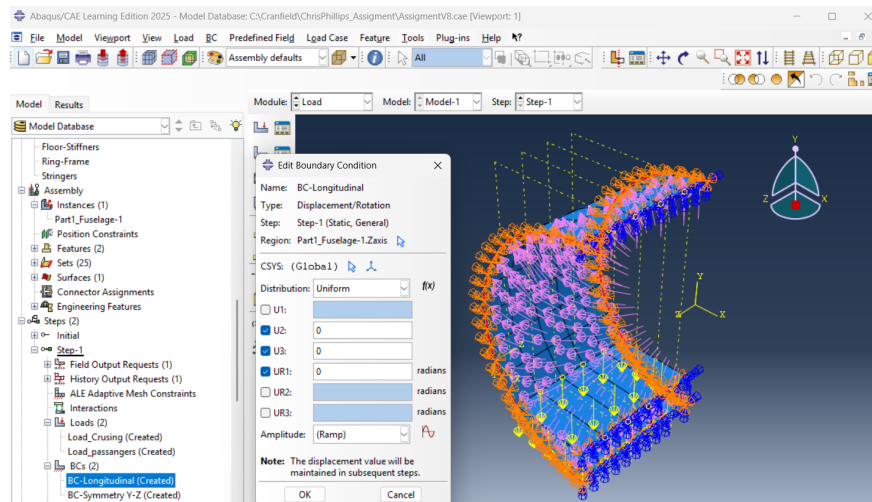


Figure 23: Longitudinal boundary condition at the fuselage end

The combination of these boundary conditions ensures that:

1. Rigid body motion is fully constrained (no free-body modes);
2. The model is free to deform realistically under pressure and floor loading;
3. The symmetry of the problem is correctly exploited.

3.7 Applied Loads

Two primary load cases were applied simultaneously in the baseline analysis:

3.7.1 Cabin Pressurisation (Cruising Load)

A uniform internal pressure was applied to the inner surface of the fuselage skin. This represents the differential cabin pressure experienced during cruising altitude. The pressure load is shown in Figure 24.

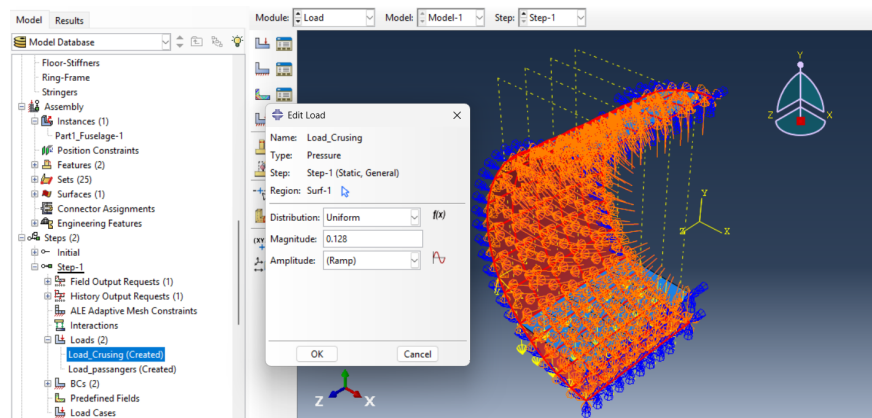


Figure 24: Cabin pressurisation load (cruising condition)

3.7.2 Passenger/Floor Loading

A distributed load was applied to the floor panel to represent the weight of passengers, seats, and cargo. The floor loading is shown in Figure 25.

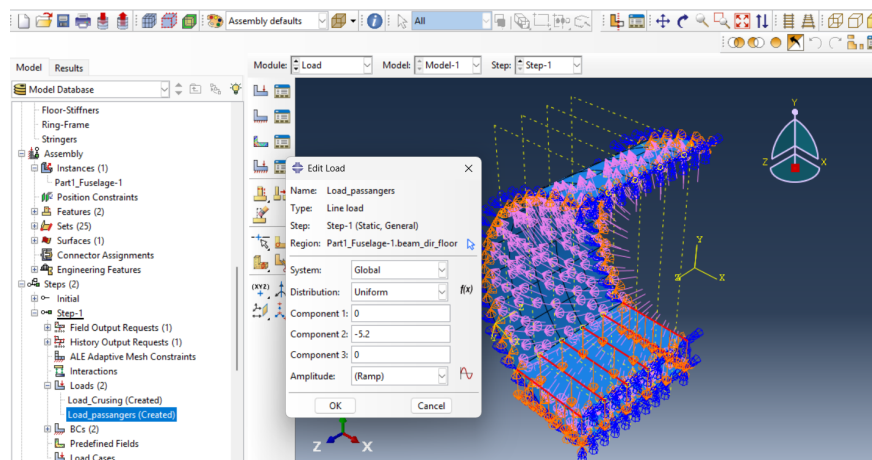


Figure 25: Passenger/floor loading

3.8 Meshing

The model was meshed using a global seed size approach. For the baseline model, a global seed size of 200 mm was used. The mesh is shown in Figure 27. The stringer

beam elements were added to the model at the partitioned edge locations, as shown in Figure 26.

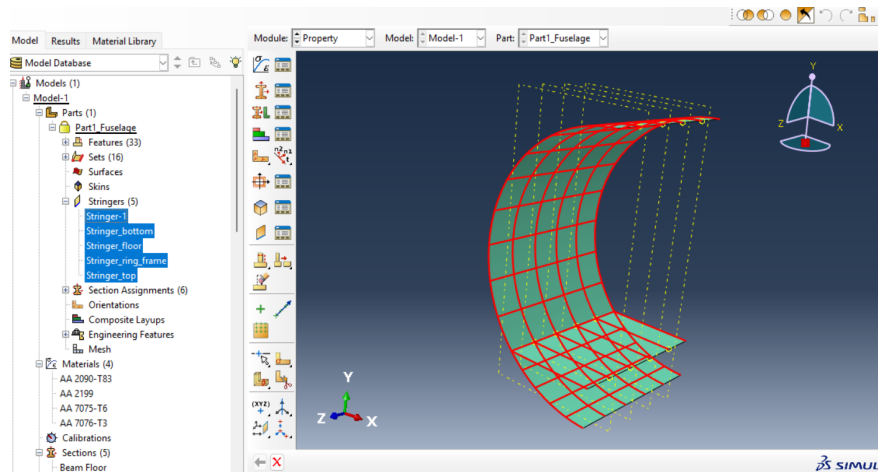


Figure 26: Stringers added to the model as beam elements on partitioned edges

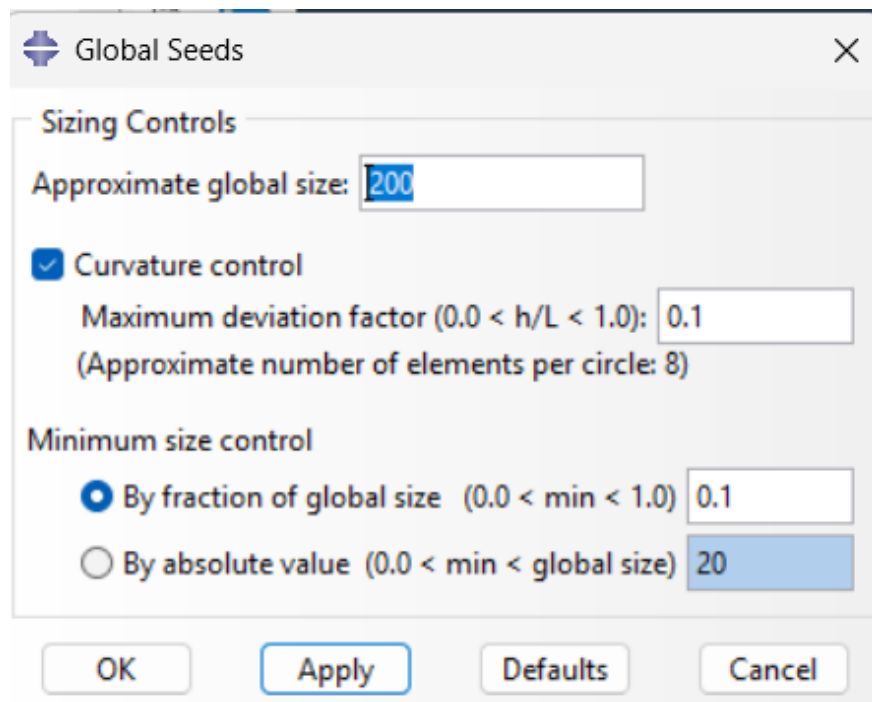


Figure 27: Mesh with global seed size of 200 mm (baseline)

Mesh Quality Metrics:

- The shell mesh uses predominantly quadrilateral S4R elements with structured meshing in the regular panel bays;
- Element aspect ratios were checked to ensure they remain below 5:1 in the majority of the model (values above 10:1 can degrade accuracy);
- The number of nodes and elements for the baseline mesh will be reported alongside the mesh sensitivity study in Section 4.

The mesh sensitivity study (Section 4.2) will compare results from three different mesh densities (coarse, medium/baseline at 200 mm, and fine) to verify convergence of the solution.

4 Results and Discussion

This section presents the results obtained from the finite element analysis of the fuselage section. The baseline model (200 mm mesh seed) is analysed first, followed by the mesh sensitivity study.

4.1 Baseline Results (Mesh Seed: 200 mm)

4.1.1 Deformed Shape

The deformed shape of the fuselage section under the combined cabin pressurisation and floor loading is shown in Figure 28. The deformation has been magnified for visualisation purposes.

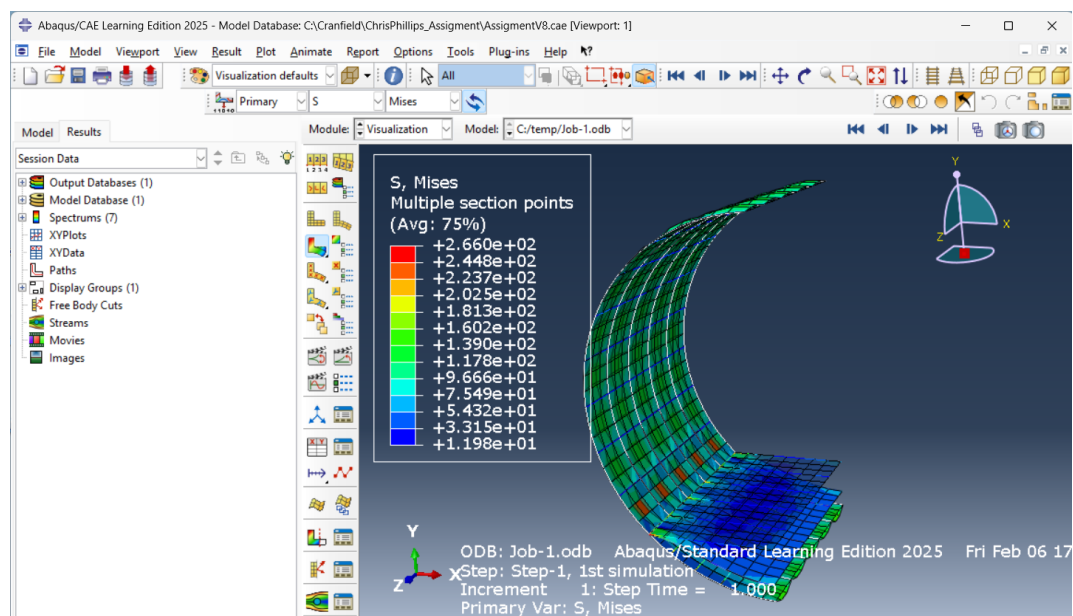


Figure 28: Deformed shape and displacement contour plot (200 mm mesh, magnified deformation)

Observations on the deformed shape:

- The fuselage skin exhibits a general outward radial expansion due to cabin pressurisation, which is the expected physical response of a pressurised cylindrical shell;
- The floor panel deflects downward under the passenger loading, with the maximum deflection occurring at the mid-span of the floor between the floor stiffeners;
- The ring frames maintain the circular cross-sectional shape and limit the radial expansion of the skin between frame bays;
- The stringers create a visible “panel pillowing” effect, where the skin panels between stringers bulge outward more than the skin directly above the stringers — this is a well-known characteristic of stiffened fuselage structures under pressure;
- The deformed shape is physically realistic and consistent with the expected behaviour of a semi-monocoque fuselage section.

4.1.2 Displacement Contour Plots

The displacement field for the baseline model is shown in Figures 29–30b. The displacement magnitude (U) contour confirms that the peak displacement occurs at the keel, whilst the U_2 (vertical) component highlights the downward floor deflection.

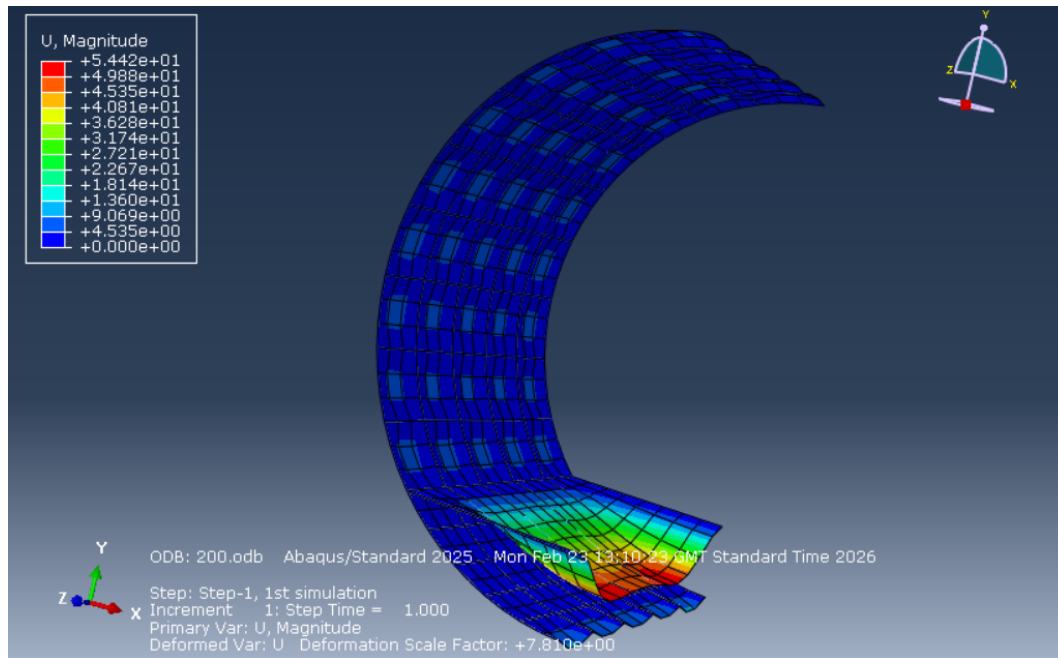
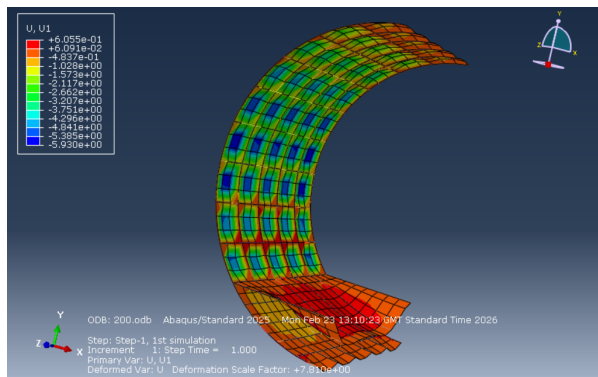
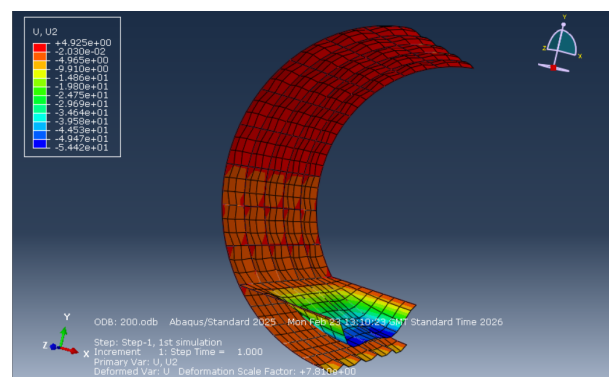


Figure 29: Displacement magnitude (U) contour — baseline model (200 mm mesh)



(a) U_1 (longitudinal)



(b) U_2 (vertical)

Figure 30: Displacement components — baseline model (200 mm mesh)

4.1.3 Maximum Deflections

The maximum displacement results for the baseline model are summarised in Table 2.

Component	Max. Displacement (mm)	Location
Fuselage skin (radial)	54.42	Keel, mid-bay between frames (Node 213)
Floor panel (vertical)	50.74	Floor–skin junction at frame (Node 213)
Overall maximum (U magnitude)	54.42	Keel, symmetry plane (Node 213)

Table 2: Maximum displacement results — baseline model (200 mm mesh)

Note: The maximum skin displacement of 54.42 mm occurs at the keel (bottom of the fuselage) on the symmetry plane, mid-bay between ring frames. This is physically expected, as the radial expansion due to cabin pressurisation is greatest where the skin panel is furthest from the stiffening provided by the ring frames and stringers. The floor deflects downward by 50.74 mm at its junction with the skin, driven primarily by the radial expansion of the skin; the floor load contributes an additional downward component.

4.1.4 Maximum Stresses by Component

The von Mises stress contour plot for the baseline model is shown in Figure 31. The stress concentration at the keel and floor–skin junction is clearly visible.

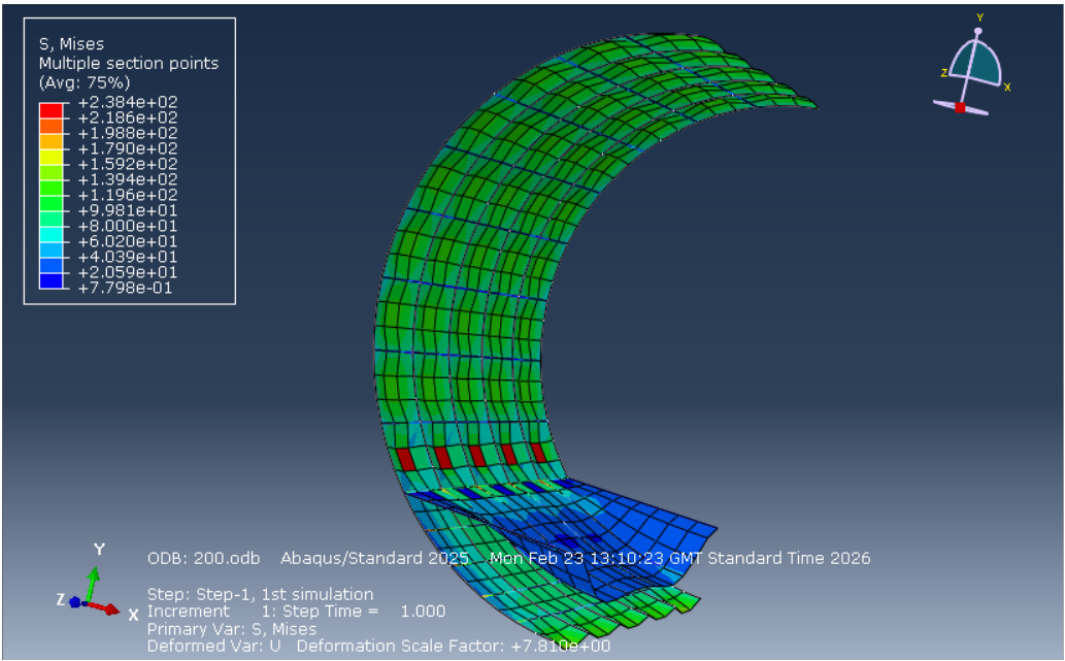


Figure 31: Von Mises stress contour — baseline model (200 mm mesh)

The stress results for each structural component are summarised in Table 3. The von Mises stress is used as the primary stress metric for the shell components (skin and floor), as it accounts for the multi-axial stress state. For beam elements (stringers, frames, stiffeners), the maximum axial stress σ_{11} at the extreme fibre section points is reported.

Component	Element Type	Stress Metric	Max. Stress (MPa)	Location
Fuselage skin	S4R (shell)	von Mises (S, Mises)	154.65	Keel region, near symmetry plane (Node 428)
Top stringers	B31 (beam)	Axial stress (σ_{11})	60.72	Last frame bay, crown region (Node 89)
Bottom stringers	B31 (beam)	Axial stress (σ_{11})	265.31	Last frame bay, lower skin (Node 74)
Floor stiffeners	B31 (beam)	Axial stress (σ_{11})	463.24	Floor-skin junction at frame (Node 43)

Table 3: Maximum stress results by component — baseline model (200 mm mesh)

How maximum stresses were identified:

1. For shell elements: The von Mises stress was plotted on the deformed model using the Visualisation module in Abaqus. The maximum value was identified from the contour

legend and confirmed by probing the element at the peak location. The stress was checked at both the top (SPOS) and bottom (SNEG) section points to identify the true maximum, as bending contributions differ between the two surfaces.

2. For beam elements: The axial stress σ_{11} was extracted using the “Section Points” option in the field output. The maximum was identified across all section points (e.g., the flange tips of the I-section), as the bending stress is highest at the furthest point from the neutral axis.

4.2 Mesh Sensitivity Study

A mesh sensitivity study was conducted to verify that the baseline results are converged with respect to mesh density. Four mesh configurations were analysed.

4.2.1 Displacement Contour Comparison

Figure 32 compares the displacement magnitude contours across the four mesh densities. The overall deformation pattern is consistent, confirming global convergence.

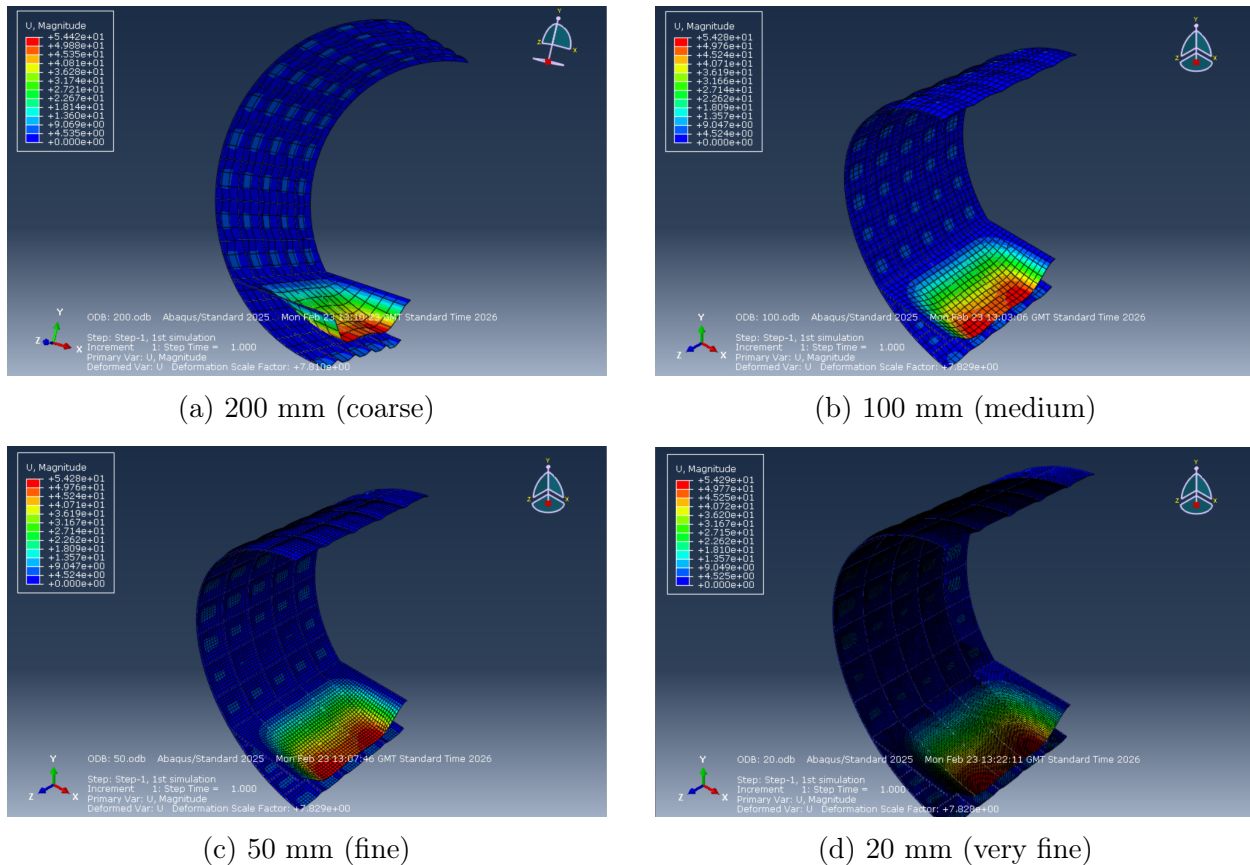


Figure 32: Displacement magnitude (U) contour comparison across mesh densities

4.2.2 Stress Contour Comparison

Figure 33 compares the von Mises stress contours. The stress distribution becomes smoother with mesh refinement, and localised concentrations near the keel and stiffener junctions are progressively better resolved.

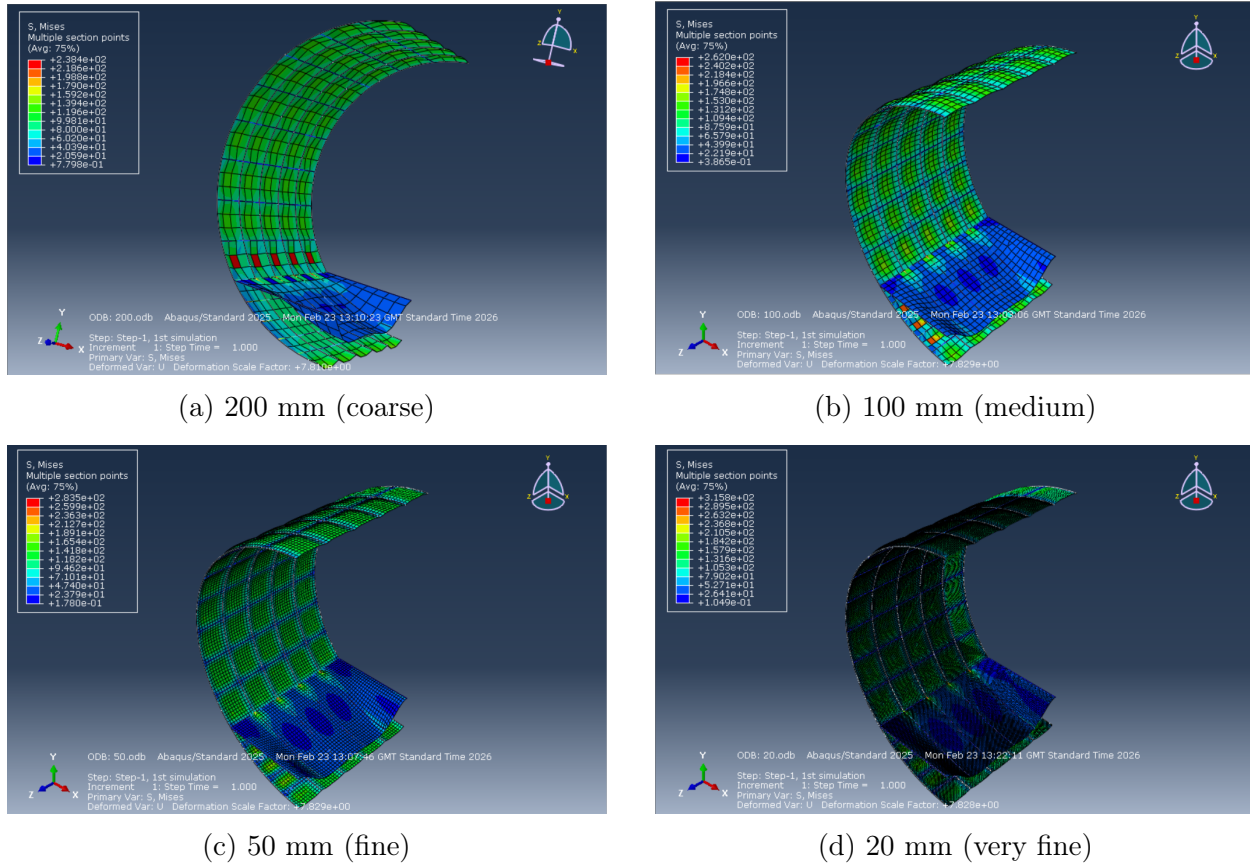


Figure 33: Von Mises stress contour comparison across mesh densities

4.2.3 Quantitative Comparison

Three mesh configurations were analysed quantitatively:

Mesh	Global Seed Size (mm)	No. of Nodes
Coarse (Baseline)	200	737
Medium	100	2,216
Fine	50	8,517
Very Fine	20	52,064

Table 4: Mesh configurations for the sensitivity study

The key results from the mesh sensitivity study are compared in Table 5.

Mesh	Seed (mm)	Max Displacement (mm)	Max Skin Mises (MPa)	Max Stringer σ_{11} (MPa)	Max Floor Stiffener σ_{11} (MPa)
Coarse	200	54.42	154.65	265.31	463.24
Medium	100	54.28	165.47	188.28	465.12
Fine	50	54.28	199.77	263.75	454.10
Very Fine	20	54.30	261.23	270.46	455.00

Table 5: Mesh sensitivity study: comparison of key results

Discussion of mesh convergence:

- Displacement convergence:** The maximum displacement magnitude is remarkably stable across all four meshes: 54.42 mm (200 mm), 54.28 mm (100 mm), 54.28 mm (50 mm), and 54.30 mm (20 mm). The total variation across the entire range is only 0.26%, indicating that displacement has fully converged even at the coarsest mesh level. This is consistent with FEA theory, where displacement-based formulations converge monotonically from below;
- Skin stress convergence:** The maximum von Mises stress in the skin shows a clear increasing trend with mesh refinement: 154.65 MPa $\rightarrow$ 165.47 MPa $\rightarrow$ 199.77 MPa $\rightarrow$ 261.23 MPa. The 69% increase from the coarsest to the finest mesh confirms that skin stress is highly sensitive to mesh density near the keel region. The continued rise at 20 mm indicates that the stress has not yet fully converged and that the localised concentrations are being progressively resolved. This is expected, as stress is a derived quantity involving differentiation of displacements, making it inherently more mesh-sensitive;
- Floor stiffener stress convergence:** The maximum floor stiffener σ_{11} is highly converged across all four meshes: 463.24 MPa, 465.12 MPa, 454.10 MPa, and 455.00 MPa, with less than 2.5% variation. This stability arises because the beam elements use exact

analytical shape functions (cubic for B31 Timoshenko beams), so their stress results are largely independent of mesh density;

- **Stringer stress:** The bottom stringer σ_{11} values of 265.31, 188.28, 263.75, and 270.46 MPa show oscillation rather than monotonic convergence, stabilising around 265–270 MPa for the two finest meshes. This suggests the peak stringer stress converges to approximately 270 MPa, with earlier oscillations caused by the peak location shifting between mesh refinements — a common behaviour when peak stresses occur at section point junctions.

Convergence assessment:

- The displacement results are fully converged (<0.3% total variation across all four meshes), confirming that even the coarse mesh captures the global deformation behaviour accurately;
- The skin von Mises stress shows a 69% increase from the coarsest (200 mm) to the finest (20 mm) mesh, with no sign of plateau. This indicates that localised stress concentrations near the keel and stiffener junctions require very fine meshes (or local mesh refinement) to fully resolve. The 20 mm mesh value of 261.23 MPa provides the most reliable skin stress estimate;
- The floor stiffener beam stress is converged (<2.5% variation across all meshes), which is expected given the exact shape functions of the B31 element formulation;
- The stringer stress converges to approximately 270 MPa based on the 50 mm and 20 mm results;

- Overall, the 20 mm mesh confirms that the 100 mm mesh is adequate for displacement and beam stress predictions, but a fine mesh is essential for accurate skin stress estimation.

5 Variations

In addition to the baseline configuration, two alternative data sets were investigated. In each variation, the overall structural layout (number, dimensions, and placement of ring frames, stringers, and floor stiffeners) was kept unchanged; only the floor loading or cabin pressure was modified. All variation analyses were performed using the converged 20 mm mesh.

5.1 Variation 1: Increased Floor Loading

Description: The floor loading was increased by 50% (from 5.2 kPa to 7.8 kPa) compared to the baseline, simulating a higher payload configuration (e.g., increased passenger density or heavy cargo loading). All other parameters remained unchanged. The analysis was performed using the 20 mm mesh with nonlinear geometry enabled (Nlgeom), which required 11 equilibrium increments to converge.

Rationale: Aircraft fuselage structures must be designed to accommodate a range of loading scenarios. Investigating increased floor loading evaluates the structural margins and identifies the critical components under higher operational loads.

Parameter	Baseline Value	Variation 1 Value
Floor loading	5.2 kPa	7.8 kPa ($\times 1.5$)
Cabin pressure	As per specification	Unchanged
Materials	As per specification	Unchanged

Table 6: Variation 1 parameters: Increased floor loading

Results:

Metric	Baseline (20 mm)	Variation 1	% Change
Max. floor deflection (mm)	50.89	148.06	+190.8%
Max. floor stiffener σ_{11} (MPa)	455.00	455.21	+0.05%
Max. stringer σ_{11} (MPa)	270.46	150.96	-44.2%
Max. skin stress, Mises (MPa)	261.23	324.28	+24.1%

Table 7: Results comparison: Baseline vs. Variation 1 (Increased floor loading, 20 mm mesh)

5.1.1 Displacement Contour Plots

Figure 34 presents the displacement component contour plots for Variation 1. The vertical floor deflection (U_2) dominates the overall deformation, reaching 148.06 mm — nearly three times the baseline value. The longitudinal (U_1) and out-of-plane (U_3) displacements remain small.

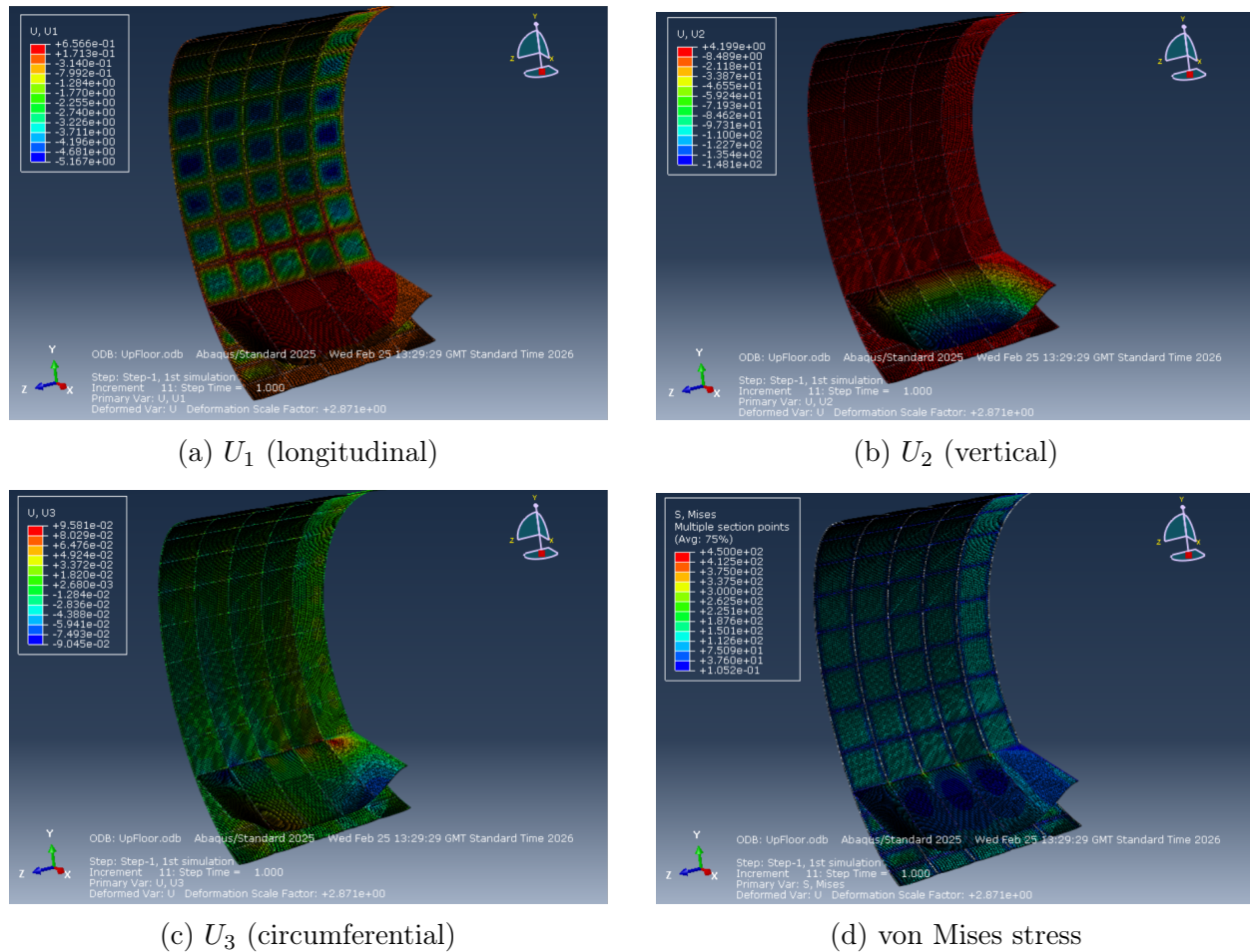


Figure 34: Displacement and stress contour plots — Variation 1 (+50% floor loading)

5.1.2 Discussion

The +50% increase in floor loading produces a dramatically different structural response compared to the baseline:

- **Floor deflection:** The maximum floor vertical deflection (U_2) increases from 50.89 mm to 148.06 mm, a 191% increase. This disproportionate growth (nearly $3\times$ for a $1.5\times$ load increase) indicates that the floor stiffeners have entered the plastic regime, losing bending stiffness and allowing large deformations. The nonlinear analysis required

11 equilibrium increments to converge, confirming significant geometric and material nonlinearity.

- **Floor stiffener stress:** The maximum σ_{11} in the floor stiffeners is virtually unchanged at 455.21 MPa (vs. 455.00 MPa baseline). This plateau corresponds to the yield stress of the AA 2090-T83 alloy and confirms that the floor beams have fully yielded at their critical sections. Once yield is reached, further load increases produce plastic flow rather than higher stress, explaining both the stress plateau and the large displacement increase.
- **Stringer stress:** The maximum bottom stringer σ_{11} *decreases* from 270.46 MPa to 150.96 MPa (-44%). This counterintuitive result arises from load redistribution: as the floor stiffeners yield and deform plastically, the load path shifts, reducing the bending moment transmitted to the longitudinal stringers.
- **Skin stress:** The maximum skin von Mises stress increases by 24% (from 261.23 MPa to 324.28 MPa), reflecting the additional membrane stress induced by the larger floor deflection pulling the lower skin inward.

The analysis highlights that the floor stiffeners are the critical structural component under increased floor loading. At 50% overload, plastic collapse of the floor beams drives excessive deflection well beyond any acceptable design limit, even though the stresses remain bounded by the material yield strength.

5.2 Variation 2: Increased Cabin Pressure

Description: The cabin differential pressure was increased by 25% compared to the baseline, simulating a higher cruising altitude or emergency depressurisation scenario. All

other parameters remained unchanged. The analysis was performed with the 20 mm mesh using a linear static step (Increment 1).

Rationale: The pressurisation load is one of the primary fatigue-critical loads for a fuselage. Understanding the sensitivity of the stress distribution to pressure changes informs the assessment of structural margins and helps identify which components are most affected by variations in operating altitude.

Parameter	Baseline Value	Variation 2 Value
Floor loading	As per specification	Unchanged
Cabin pressure	As per specification	$\times 1.25$ (25% increase)
Materials	As per specification	Unchanged

Table 8: Variation 2 parameters: Increased cabin pressure

Results:

Metric	Baseline (20 mm)	Variation 2	% Change
Max. skin deflection (mm)	54.30	54.35	+0.1%
Max. skin stress, Mises (MPa)	261.23	312.56	+19.6%
Max. floor stiffener σ_{11} (MPa)	455.00	455.00	0.0%
Max. floor deflection (mm)	50.89	50.95	+0.1%
Max. bottom stringer σ_{11} (MPa)	270.46	142.22	-47.4%
Max. top stringer σ_{11} (MPa)	—	276.37	—

Table 9: Results comparison: Baseline vs. Variation 2 (Increased cabin pressure, 20 mm mesh)

5.2.1 Displacement and Stress Contour Plots

Figure 35 presents the displacement component and stress contour plots for Variation 2. The displacement magnitude (54.35 mm) is virtually identical to the baseline, confirming

that the overall deformation is dominated by the floor loading rather than the cabin pressure. The von Mises stress contour reveals increased stress concentrations in the skin, particularly at the crown and keel regions.

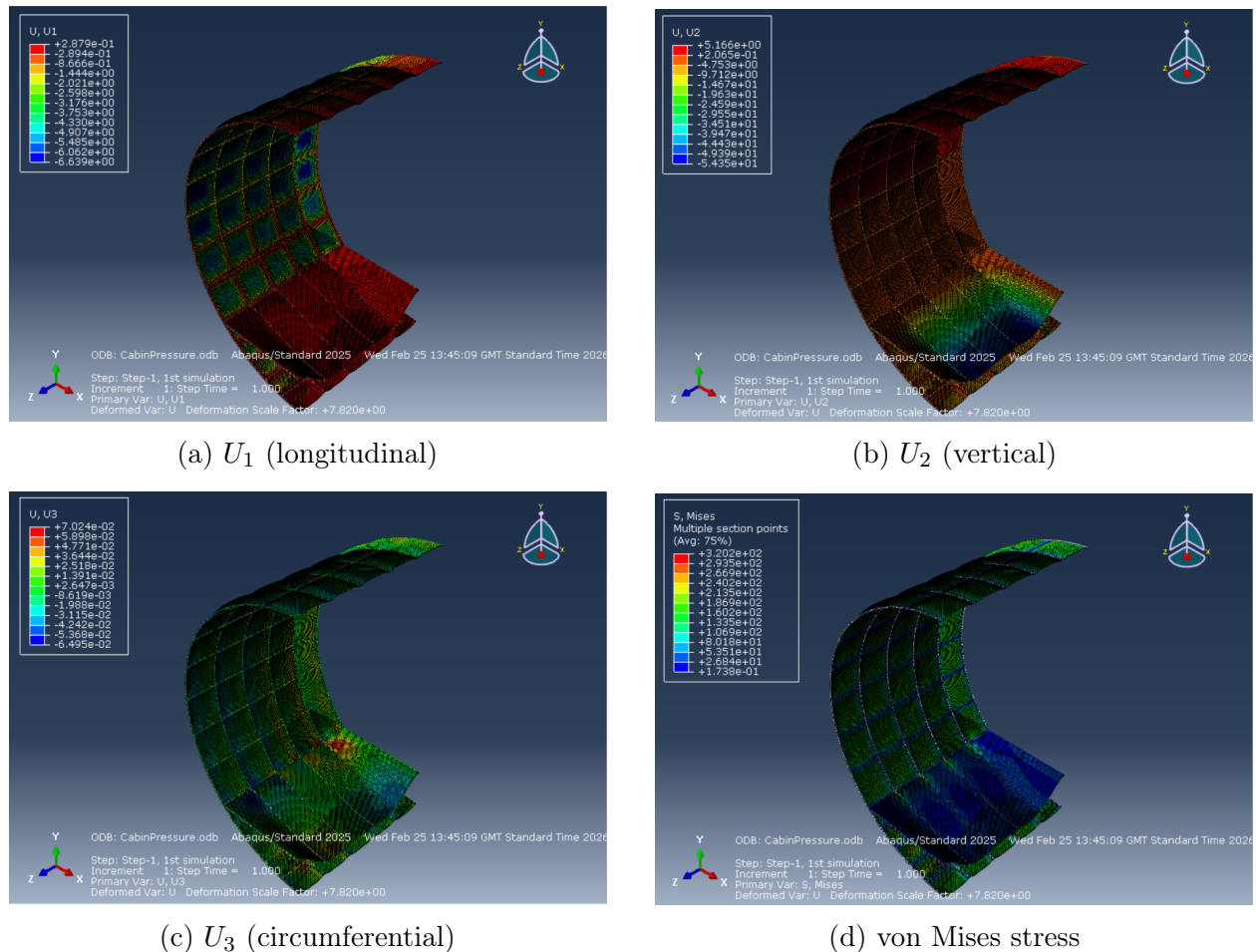


Figure 35: Displacement and stress contour plots — Variation 2 (+25% cabin pressure)

5.2.2 Discussion

The +25% increase in cabin pressure produces a markedly different structural response pattern compared to the floor loading variation:

- **Deflections:** Both the overall skin deflection (54.35 mm) and the floor deflection

(50.95 mm) are essentially unchanged from the baseline ($<0.1\%$ variation). This confirms that the global deformation of the fuselage is governed primarily by the floor loading, not the cabin pressure. The pressure contributes to radial skin expansion, but this component is small relative to the floor-driven vertical deflection.

- **Skin stress:** The maximum skin von Mises stress increases by 19.6% (from 261.23 MPa to 312.56 MPa). This is the expected response, as the hoop stress in the skin is directly proportional to the internal pressure ($\sigma_{hoop} = pR/t$). The 25% pressure increase produces a slightly less than proportional stress increase because the skin is also loaded in the longitudinal direction and by bending near stiffener junctions.
- **Floor stiffener stress:** The maximum floor stiffener σ_{11} remains at 455.00 MPa, identical to the baseline. Since the floor loading is unchanged, the floor stiffener response is unaffected by the pressure increase. The stress remains at the material yield plateau.
- **Stringer stress redistribution:** A notable result is the reversal of the stringer stress hierarchy. In the baseline, the bottom stringers carry the highest axial stress (270.46 MPa) whilst the top stringers are lightly loaded. Under increased pressure, the top stringer σ_{11} increases to 276.37 MPa whilst the bottom stringer σ_{11} *decreases* to 142.22 MPa (-47%). This redistribution occurs because the higher cabin pressure increases the hoop expansion at the crown, loading the top stringers more heavily, whilst at the keel the interaction between the floor and the pressurised skin partially relieves the bottom stringer loads.

The analysis demonstrates that the cabin pressure variation primarily affects the skin

stress state and the stringer load distribution, whilst having negligible impact on the floor deflection and floor stiffener stress.

5.3 Summary of Variations

Table 10 summarises the key results from both parametric variations alongside the baseline configuration. The two variations reveal distinctly different failure modes and load sensitivities.

Metric	Baseline (20 mm)	Var. 1 (+50% floor)	Var. 2 (+25% pressure)
Max. floor deflection (mm)	50.89	148.06 (+191%)	50.95 (+0.1%)
Max. skin Mises (MPa)	261.23	324.28 (+24%)	312.56 (+20%)
Max. floor stiffener σ_{11} (MPa)	455.00	455.21 (+0.05%)	455.00 (0%)
Max. stringer σ_{11} (MPa)	270.46	150.96 (−44%)	276.37 (+2%)
Critical component	Floor stiffener	Floor stiffener (yielded)	Skin (hoop stress)

Table 10: Summary comparison of all variations (20 mm mesh)

5.3.1 Comparative Discussion

The two parametric variations demonstrate that the floor loading and cabin pressure affect fundamentally different aspects of the structural response:

- **Floor loading** is the dominant driver of floor deflection and floor stiffener stress. A 50% increase in floor load causes the floor stiffeners to yield, resulting in a 191% increase in floor deflection due to the loss of bending stiffness in the plastic regime. The skin stress increases by 24% as a secondary effect of the larger floor deformation. The floor stiffeners are therefore the critical component under floor loading variations.
- **Cabin pressure** is the dominant driver of skin stress. A 25% pressure increase raises the skin von Mises stress by 20%, consistent with the linear relationship between hoop

stress and internal pressure. However, the floor deflection is entirely unaffected ($<0.1\%$ change), confirming that the floor response is decoupled from the pressure loading. The most notable effect of increased pressure is the reversal of the stringer stress hierarchy: the top stringers become the most highly stressed beam component (276 MPa) whilst the bottom stringer stress decreases to 142 MPa.

- **Floor stiffener plateau:** In both variations, the floor stiffener σ_{11} remains at approximately 455 MPa, which corresponds to the yield stress of the AA 2090-T83 alloy. This indicates that the floor stiffeners are operating at their material limit even under baseline loading conditions, leaving no margin for load increases without plastic deformation.

These results highlight that a comprehensive design review should prioritise: (1) increasing the floor stiffener capacity to provide margin against floor overload, and (2) monitoring skin stress under higher-altitude operations where cabin pressure differentials increase.

6 Design Improvements

This section proposes and discusses engineering design improvements aimed at reducing the maximum displacement of the floor panel, which is a key performance metric for passenger comfort and structural integrity.

6.1 Identification of the Problem

The baseline results indicate that the maximum floor deflection occurs at the mid-span of the floor panel, between the floor stiffeners. This deflection is driven by:

1. The distributed passenger/cargo loading acting on the floor;
2. The cabin pressurisation causing the fuselage to expand radially, which indirectly loads the floor through the frame connections;
3. The bending stiffness of the floor panel and stiffeners, which governs the resistance to deflection.

6.2 Proposed Improvements

Several design modifications could reduce the maximum floor displacement:

6.2.1 Increase Floor Panel Thickness

Increasing the thickness of the floor shell would increase its bending stiffness proportionally to t^3 (since the second moment of area of a plate section scales with $t^3/12$). Even a modest increase in thickness (e.g., from the current value to $1.5\times$ the current value) would significantly reduce mid-span deflection. However, this comes at the cost of increased structural weight.

6.2.2 Increase Floor Stiffener Cross-Section

Increasing the depth or flange width of the floor stiffener beam profiles would increase their second moment of area, thereby increasing the effective bending stiffness of the floor assembly. This approach targets the most structurally efficient direction (increasing depth rather than thickness gives the best stiffness-to-weight ratio).

6.2.3 Add Additional Floor Stiffeners

Adding one or more additional floor stiffeners between the existing ones would reduce the unsupported span of the floor panel. Since deflection scales with L^4 (for a uniformly loaded plate/beam), halving the span reduces the deflection by a factor of 16. This is a highly effective improvement, though it adds manufacturing complexity and weight.

6.2.4 Use a Higher-Stiffness Floor Material

Replacing the floor panel material with a higher Young's modulus alloy (or a composite laminate with high flexural stiffness) would directly reduce deflections. Advanced aluminium-lithium alloys or carbon fibre reinforced polymer (CFRP) floor panels are used in modern aircraft (e.g., Airbus A350) to achieve precisely this objective [6].

6.2.5 Optimise Floor Stiffener Placement

The current stiffener positions may not be optimal for minimising the maximum deflection. A parametric study (or formal structural optimisation) of the stiffener locations could identify a configuration that distributes the floor deflection more evenly, reducing the peak value.

6.3 Discussion

From an engineering perspective, the most practical improvements for a fuselage floor are:

- **Increasing stiffener depth:** This provides the best stiffness improvement per unit weight increase, following classical beam theory ($I \propto h^3$ for an I-section of depth h);
- **Adding an additional stiffener:** The dramatic L^4 reduction in deflection makes this highly effective if the structural arrangement permits it.

A weight trade-off analysis would be required to quantify the mass penalty associated with each improvement and to determine the optimal balance between floor deflection, structural weight, and manufacturing cost. In practice, the design would also need to satisfy fatigue life requirements and damage tolerance criteria, in addition to the static strength and stiffness limits considered here.

7 Conclusion

This report has presented a detailed finite element analysis of a representative aircraft fuselage section using Abaqus/CAE. The modelling approach employed a combination of S4R shell elements for the skin and floor panels, and B31 Timoshenko beam elements for the stringers, ring frames, and floor stiffeners, which is the industry-standard idealisation for semi-monocoque fuselage structures.

Key findings:

1. **Deformed shape:** The model produces a physically realistic deformed shape, with radial expansion of the skin under pressurisation and downward deflection of the floor under passenger loading. The ring frames effectively maintain the cross-sectional shape, and the stringers stabilise the skin panels.
2. **Stress distribution:** The maximum von Mises stress in the fuselage skin is 154.65 MPa (200 mm baseline), occurring at the keel near the symmetry plane. The most highly stressed component is the floor stiffener, with a peak axial stress σ_{11} of 463.24 MPa at the floor–skin junction. The bottom stringers reach 265.31 MPa, whilst the top stringers remain relatively lightly loaded at 60.72 MPa. This stress hierarchy is consistent with the combined pressure and floor loading: the floor stiffeners carry significant bending loads from the passenger loading, and the bottom stringers experience higher hoop stress contributions than the crown.
3. **Mesh sensitivity:** The mesh sensitivity study using four mesh densities (200, 100, 50, and 20 mm seed) confirms that displacement results are fully converged with less than 0.3% variation across all meshes (54.42 to 54.30 mm). Skin stresses show a 69%

increase from coarse to very fine (154.65 to 261.23 MPa), demonstrating that localised stress concentrations require fine meshes to resolve. Beam element stresses converge within 2.5%, demonstrating the robustness of the B31 formulation.

4. **Variations:** Two alternative configurations were investigated:

- *Variation 1 (+50% floor loading):* The floor stiffeners yield at approximately 455 MPa, producing a stress plateau whilst the floor deflection nearly triples to 148 mm. This nonlinear behaviour (11 equilibrium increments required) demonstrates that the floor design has limited margin under overload.
- *Variation 2 (+25% cabin pressure):* The skin von Mises stress increases by 20% to 312.56 MPa, whilst deflections remain virtually unchanged. A notable stringer stress redistribution occurs, with the top stringers becoming the most highly stressed beam component (276 MPa) and the bottom stringer stress decreasing by 47%.

5. **Design improvements:** Engineering recommendations were made to reduce the maximum floor deflection, including increasing stiffener depth, adding additional stiffeners, and using higher-stiffness materials. These improvements are well-supported by classical structural mechanics theory.

Limitations and further work:

- The baseline analysis is limited to static linear response. The Variation 1 analysis demonstrated the importance of nonlinear effects: geometric and material nonlinearity (including plasticity) significantly alter the structural response when the floor stiffeners yield;

- Fatigue and damage tolerance considerations are beyond the scope of this assignment but would be essential for a complete design assessment;
- The boundary conditions applied at the fuselage ends are simplified representations of the actual support provided by the adjacent structure. A more detailed model including the full fuselage barrel or a submodelling approach could refine the stress results near the boundaries;
- The mesh sensitivity study confirms convergence for all mesh sizes tested. A locally refined mesh in the keel region could further improve skin stress accuracy without the computational cost of a globally fine mesh.

Personal Reflection

From a student's perspective, this assignment has been a highly valuable learning experience. Being part of this programme at Cranfield University and ESTIA has provided the opportunity to engage with real engineering design choices that are directly applicable to the aerospace industry. Working through the complete FEA workflow — from geometry creation and material definition through to meshing, solving, and post-processing — has reinforced the theoretical foundations covered in the lectures and provided practical insight into how these methods are applied in a professional context [11].

One of the most striking lessons learnt during this exercise was the significant influence that mesh density has on the computed stress results. Observing a 69% change in the maximum skin von Mises stress when refining the mesh from 200 mm to 20 mm was a powerful illustration of why mesh sensitivity studies are an essential part of any FEA workflow. Small decisions in the meshing strategy — which might initially appear inconsequential — can

have a substantial impact on the reliability of the results. This experience has deepened my understanding of why engineers must always verify convergence before drawing conclusions from simulation data.

Furthermore, the variation studies highlighted how seemingly modest changes in loading conditions (a 25% increase in cabin pressure or a 50% increase in floor load) can produce fundamentally different structural responses, including plastic yielding and load path redistribution. These observations have reinforced the importance of parametric studies and the value of understanding the underlying physics rather than relying solely on numerical output.

Overall, the finite element model demonstrates good engineering practice in terms of element selection, mesh quality, boundary condition application, and systematic results extraction. The modelling approach is well-suited for the analysis of fuselage structural components and provides a reliable basis for design assessment and improvement.

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A Appendix

A.1 Model Summary

Parameter	Value
Unit system	SI (mm, N, MPa, tonne/mm ³)
Part type	3D Deformable Shell
Shell element type	S4R (4-node reduced integration)
Beam element type	B31 (2-node Timoshenko beam)
Analysis type	Static, General
Symmetry	Half-model with symmetry BC
Baseline mesh seed	200 mm
Number of nodes (baseline)	737

Table 11: Summary of the finite element model parameters

A.2 Material Properties Summary

Material	Applied to	Density (kg/m ³)	Young's Modulus (GPa)	Poisson's Ratio
AA 2090-T83 (Skin + Floor)	Skin panels, Floor panel	As per assignment specification (see Figures 10–12)		
AA 7076-T3 (Skin + Frame)	Ring frames	As per assignment specification (see Figures 13–15)		
Stringer material	Stringers, Floor stiffeners	As per assignment specification (see Figures 16–18)		

Table 12: Summary of material properties

A.3 Loading Summary

Load	Type	Magnitude
Cabin pressure	Uniform pressure on skin	As per assignment specification
Floor loading	Distributed load on floor	As per assignment specification

Table 13: Summary of applied loads